No of Pages : 3 Course Code : 23MX13

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, DECEMBER 2023

MCA Semester : 1

23MX13 DATA STRUCTURES

Time : 3 Hours **Maximum Marks : 100**

INSTRUCTIONS:

1. Answer **ALL** questions. Each question carries 25 Marks.
2. Subdivision (a) carries 5 marks each, subdivision (b) carries 8 marks each and subdivision (c) carries 12 marks each.
3. If any data is found missing, suitable assumptions can be made with proper justification.
4. Course Outcome : Table

Qn.1	CO1	Qn.2	CO2	Qn.3	CO3	Qn.4	CO4

1. a) i) What is an ADT? What is its need? Define Queue ADT. (5)

b) What are the characteristics properties of an algorithm? Write an algorithm for printing the first n elements in the Fibonacci sequence. Show that your algorithm possess all the properties of an algorithm. (8)

c) i) Indicate the complexity of the following code segments using Big O notation. (4)

```
1) for ( j = 1; j ≤ n; j++)
      for( k = n; k > 1; k = k/2)
         for (m =1; m ≤ n ; m = m *2)
            printf("$")
```

```
2) for(j = 1; j < n; j++)
      for( k = 1 ; k < n ; k++) {
         print("$")
         break;
         }
```

ii) Define Big O notation. Write the significance of Big O notation. Solve the following recurrence relation, using substitution method and write the complexity using Big O notation (8)

$$T(n) = T(n-1) + T(n-2) + c$$

$$T(1) = 1$$

2. a) What are the various ways of representing two dimensional matrix in storage? Explain with suitable examples. Write the corresponding addressing functions. Suggest an efficient representation for storing an upper triangular matrix and write the addressing function. (5)

b) i) Write an algorithm using Stack ADT calls to recognize strings with balanced parentheses and square brackets. Trace the algorithm with strings ([( )]), (( )]) (4)

ii) Implement a queue using a singly linked list. (4)

c) i) Consider the ADT Set that represents a collection of Integers. The ADT should support standard set operations such as the following:

i. S = init( ) /* Initialize S to the empty Set */

ii. isSingleton(S) /* Return True if and only if S is an empty Set */

iii. isMember(S,a) /* Return True if and only if a is a member of the Set S */

iv. S = addElement(S,a) /* Add a to the Set S. If a is already in S, there will no change in S */

v. S = delElement(S,a) /* Remove a from the Set S. If a is not a member of S, there will no change */

vi. Print(S) /* Print the elements of S */

Implement the Set ADT using Linked Lists (12)

(OR)

ii) Formulate algorithms for the following

1) Given two sorted list, merge them as a singly linked list

2) Given a doubly linked list and a key, print the data stored in the predecessor and successor nodes of the key and delete the key element. If the key element is not present, print an appropriate message.

3. a) Write the sequence in which nodes are traversed in inorder, preorder and postorder traversal on Binary Trees. Draw a binary tree whose inorder traversal sequence is H D J B E A F J C G and post order traversal sequence is H J D E B J F G C A. (5)

Draw a Binary Tree in which all the non-leaf nodes have two children and whose pre order traversal sequence is 100, 34, 16, 9, 8, 38, 11, 4, 81 and post order traversal sequence is 34, 9, 11, 4, 38, 81, 8, 16, 100.

b) i) Define heap data structure. Why is heap a preferred data structure to implement priority queue? Illustrate the max heap construction with the keys 34, 56, 17, 12, 8, 3, 2, 1, 0. What is the time complexity for inserting an element into the heap? (5)

ii) Draw a) directed and b) undirected graph with 5 vertices and 6 edges. Show their representation using Adjacency matrix and Adjacency list (3)

c) i) Write algorithms for Breadth First Search and Depth First Search on Graphs. Tracing the algorithms, write the traversal sequence for the graph given in fig.1. Modify any one of the algorithms to check whether the given graph is connected. (12)

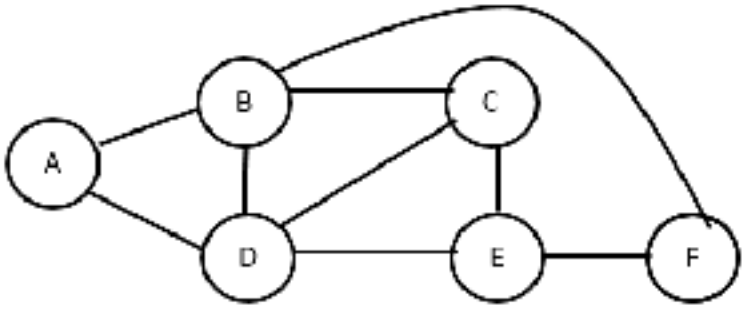

Fig.1

(OR)

ii) What is a Binary Search Tree? Formulate algorithms for insertion and deletion on a Binary Search Tree. Create a Binary Search tree of level 4 with the following elements : 45, 65, 78, 90, 23, 30, 10, 50, 5, 10, 60, 12

4. a) Illustrate heap sort and radix sort with the following numbers. 45, 65, 78, 90, 23, 30, 10, 50. Show the result of the intermediate passes. What is the complexity of these algorithms? (5)

b) i) Formulate an algorithm for binary search. When Binary Search algorithm is applied to search for keys 67 and 15 in the list of elements 2, 4, 12, 22, 37, 39, 45, 48, 52, 64, 67, 69, 80, in which order elements will be compared with the keys. What is the time complexity for successful and unsuccessful search? (4)

ii) Formulate algorithm for Bubble sort. What is the complexity of Bubble sort algorithm. (4)

c) What is Hashing? Describe any two methods for finding the hash address. What is meant by Collision in hashing? Describe any two ways of resolving collisions. The following elements are to be entered in the given order in a hash table of size 11: 12, 31, 63, 89, 90, 78, 55 and 21. Apply a suitable hash function. Show the contents of the table. Illustrate collision resolution methods described. For each of the methods, how many comparisons are to be made in locating the elements: 31, 89 and 56. (12)

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FD/RL